

Industrial CO2 Sensor with Modbus RTU Communication – Datasheet

1. Product Overview

WIN-SN-AIRV-M is a versatile data acquisition device designed for real-time monitoring in industrial applications, especially ducts. It supports a wide range of external sensors with **I²C, Analog, and UART interfaces**, providing flexibility for various measurements. The device integrates seamlessly with **Modbus RS-485 networks**, offering reliable communication and long-range data transfer over several meters, making it ideal for large-scale installations.

Its robust design makes it suitable for **installations in HVAC systems, environmental monitoring, and process control**. With **configurable data sampling rates, multi-sensor compatibility, and easy configuration**, WIN-SN-AIRV-M ensures **accurate Air velocity data acquisition and reliable performance** in demanding industrial environments, making it an excellent choice for **industrial and IOT applications**.



Fig.1 - Product Image

2. Key Technical Parameters

- Measurement Range: 0.3 to 3.0 m/s
- Accuracy: ± 0.05 m/s, $\pm 1\%$
- Detection Accuracy: $\sim 90\%$
- **RS-485 Modbus RTU** communication
- Configurable data sampling rates

Measuring Parameter	Measuring Range	Accuracy
Air Velocity	0.3 to 3.0	±0.05 m/s,

Table 1

2. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*36 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

4. Configuration Settings

- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None
- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

6. Modbus RS485 Data Storage register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	AIRV Values	40001	0x03	RS485	16-bit int
2	Display Baud Rate (Default: 960)	40011	0x03	RS485	16-bit int
3	Enter New Baud Rate	40012	0x03	RS485	16-bit int
4	Display Slave ID (Default: 1)	40013	0x03	RS485	16-bit int
5	Enter New Slave ID	40014	0x03	RS485	16-bit int
6	Display Parity (Default: None-3 And for Odd-1, Even-2)	40015	0x03	RS485	16-bit int
7	Enter New Parity	40016	0x03	RS485	16-bit int
8	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16-bit int
9	Enter New Start/ Stop Bit	40018	0x03	RS485	16-bit int

7. Modbus Configuration

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below.

Important Note

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.

Connection Setup

Connection: Serial Port OK Cancel

Serial Settings

USB-SERIAL CH340 (COM15) Mode ☒ RTU ☐ ASCII

9600 Baud Response Timeout 100 [ms]

8 Data bits Delay Between Polls 10 [ms]

None Parity Advanced...

1 Stop Bit

Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502 Connect Timeout: 1000 [ms] ☒ IPv4 ☐ IPv6

Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address mode: ☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

Scan Rate: 1000 [ms] Apply

Mbpoll1

Tx = 418: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	AIRV	33	Current Baud	960
1		0	New Baud	0
2		0	Current SID	1
3		0	New SID	0
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

- Above image Shows sensor data over Modbus with default Modbus settings
- At the register 40001 shows AIR Velocity Values.

- As you can see in the image above, the resistor 40011 – 960 represents the present baud rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.

Mbpoll1

Tx = 418: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	AIRV	33	Current Baud	960
1		0	New Baud	11520
2		0	Current SID	1
3		0	New SID	10
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

Mbpoll1

Tx = 418: Err = 4: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	AIRV	33	Current Baud	11520
1		0	New Baud	0
2		0	Current SID	10
3		0	New SID	0
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

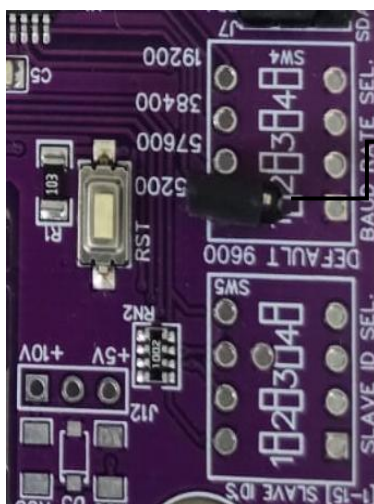


Image1.1



Image1.2

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.



Contact Information

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